Solving for the toponium Green's function with the method of Jeżabek, Kühn and Teubner

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Abstract

Artemiy's funky attempt at toponium. Near threshold, everything about a $t\bar{t}$ pair that feels the QCD potential is contained in one complex function, the S-wave Green's function $G(E, p)$. This note shows how I compute it from scratch by following [JKT] equation by equation: the potential of their Sec. 2, the regularisation and numerical solution of their Sec. 3, and the checks that the solver is right. It then puts G into MadGraph's $pp \to t\bar{t}$ matrix element, for the attractive colour singlet and the repulsive colour octet, and extracts toponium K-factors for binned $t\bar{t}$ distributions at 13 TeV. The event-level recipe is that of Fuks et al.; the octet potential follows Hagiwara, Sumino and Yokoya.

References (PDFs in `reference/`)

- [JKT] M. Jeżabek, J.H. Kühn, T. Teubner, Z. Phys. C 56 (1992) 653 (`BF01474740.pdf`). *The method.*
- [Fuks] B. Fuks, K. Hagiwara, K. Ma, Y.-J. Zheng, arXiv:2411.18962v2 (`2411.18962v2.pdf`). *The event recipe, (15)–(16), (19)–(21), and their tables of G.*
- [HSY] K. Hagiwara, Y. Sumino, H. Yokoya, arXiv:0804.1014. *The colour-octet potential, (9)–(11).*

Equation numbers always refer to the paper named in front of them: [JKT] (21) is Eq. (21) of JKT. The code is one library, `toponium.py`; a "→ `name`" at the end of a paragraph names the class or method that implements it.

1 The problem

A $t\bar{t}$ pair produced at a point with energy $E = W - 2m_t$ (W the pair's invariant mass) and relative momentum p propagates as

$$G(\vec{p}, E) = \int d^3r\, e^{i\vec{p}\cdot\vec{r}}\, G(\vec{r}, 0; E) = \left\langle \vec{p} \middle| \frac{1}{\mathcal{E} - H} \middle| \vec{r} = 0 \right\rangle,$$

$$H = \frac{p^2}{m_t} + V(r), \qquad \mathcal{E} = E + i\Gamma_t \qquad ([\text{JKT}]\ (2)\text{–}(4)).$$

The width enters through $E \to \mathcal{E}$: the tops decay before the pair can form a stable bound state, so the toponium "resonances" are smeared into a single bump. Without the potential,

$$G_0(p) = \frac{1}{\mathcal{E} - p^2/m_t} \qquad ([\text{JKT}]\ (6)),$$

and the potential's whole effect is in the ratio G/G_0. In particular, re-weighting each MadGraph $t\bar{t}$ event by $|G(E, p^*)/G_0(E, p^*)|^2$ ([Fuks] (15)) puts toponium into the event sample.

So the task is: *compute $G(E,p)$ for the QCD potential.* [JKT] give both the potential (their Sec. 2) and a numerical method (their Sec. 3), and I follow both as printed. Where they leave a choice open, I say so and list it in Sec. 8.

Relation to Fuks et al. [Fuks] use the same re-weighting but do not say how they computed G; they publish it as a table (swData) and cite [JKT]. My G is computed independently of those tables. Comparing the two turned out to be a test of the tables rather than of my solver (Sec. 6).

Conventions: $m_t = 173$ GeV, $\Gamma_t = 1.49$ GeV, $C_F = 4/3$, $n_F = 5$, $\alpha_s(m_Z) = 0.12$. G has the sign of [JKT] (6). → `Constants, Constants.G0`

2 The method at a glance

[JKT]	what it is	in `toponium.py`
(14)	two-loop running coupling $\alpha_s(Q^2)$	`JKTPotential.alpha_2loop`, `lambda_msbar_2loop`
(13)	perturbative potential $\tilde{V}_{\rm pert}(Q^2)$	`JKTPotential.V_pert`
(15)	full potential: (13) above 5 GeV, Richardson-like below	`V_ir`, `continuity_constant_C`, `V_jkt`
(19)–(20)	cut at small momentum, and its normalisation	`V_cut`, `V_cut_position`, `energy_shift`
(16)–(17)	$G = G_0 K$ and the integral equation for K	`LSSolver.G`
(18)	S-wave kernel $\hat{V}(p,q)$	`LSSolver.kernel`
(21)–(22)	subtracted equation and $\mathcal{A}(p)$	`LSSolver.script_A`, `solve_K`
(23)	Gauss–Legendre grid	`LSSolver.jkt_grid`

Algorithm. Input: $\alpha_s(m_Z)$, m_t, Γ_t.

1. Build the potential: $\Lambda_{\overline{\rm MS}}$ from (14) at m_Z, then (13), (15), the cut (19).	Sec. 3
2. Fix the cut's δ^3 term by (20); it becomes a constant energy shift.	Sec. 3
3. Tabulate $F(s) = \int_0^s V\, dk^2$, which gives the kernel (18).	Step 2
4. Nodes and weights q_j, w_j (23); matrix $W_{ij} = w_j \hat{V}(q_i, q_j)$, zero diagonal.	Steps 2, 5
5. For each E: $g_j = G_0(q_j; E - \text{shift})$, $\mathcal{A}(q_i)$ (22), solve (21) for K_i.	Steps 3, 4, 6
6. For each output p: $G(p) = G_0(p)\, K(p)$ (16), K splined between nodes.	Step 7

3 The potential (JKT Sec. 2 and the cut of Sec. 3)

Everything here is in momentum space: $\tilde{V}(Q^2)$, with Q the momentum transfer. → `JKTPotential`

The running coupling, [JKT] (14).

$$\alpha_{\overline{\rm MS}}(Q^2) = \frac{4\pi}{b_0 \ln\!\big(Q^2/\Lambda^2_{\overline{\rm MS}}\big) + \dfrac{b_1}{b_0} \ln\ln\!\big(Q^2/\Lambda^2_{\overline{\rm MS}}\big)}, \qquad b_0 = 11 - \tfrac{2}{3}n_F, \quad b_1 = 102 - \tfrac{38}{3}n_F.$$

JKT quote α_s directly; I quote $\alpha_s(m_Z) = 0.12$, the value [Fuks] use, and *solve* (14) for Λ: with $x = \ln(m_Z^2/\Lambda^2)$, (14) at $Q = m_Z$ reads

$$b_0\, x + \frac{b_1}{b_0} \ln x = \frac{4\pi}{\alpha_s(m_Z)},$$

a single equation in x, solved by root-finding (Brent's method). This is the only equation solved in building the potential; everything after it is evaluated directly. For $\alpha_s(m_Z) = 0.12$, $n_F = 5$: $\Lambda^{(5)}_{\overline{\rm MS}} = 223$ MeV, and (14) then gives $\alpha_s = 0.221,\ 0.183,\ 0.157,\ 0.150$ at $Q = 5,\ 10,\ 20,\ 25$ GeV. → `beta_coefficients, lambda_msbar_2loop, alpha_2loop`

The perturbative potential, [JKT] (13).

$$\tilde V_{\rm pert}(Q^2) = -\frac{16\pi}{3}\,\frac{\alpha_{\overline{\rm MS}}(Q^2)}{Q^2}\left[1 + \left(\frac{31}{3} - \frac{10}{9}n_F\right)\frac{\alpha_{\overline{\rm MS}}(Q^2)}{4\pi}\right].$$

This is one-gluon exchange, $-4\pi C_F\alpha/Q^2$ with $4\pi C_F = 16\pi/3$, with two improvements: the coupling runs with the momentum transfer, $\mu^2 = Q^2$, by (14), and the square bracket is the one-loop correction to the potential. Nothing is solved: for each Q^2, compute $\alpha_{\overline{\rm MS}}(Q^2)$ from (14), put it into (13). (With α held fixed this is the potential of [Fuks] (12); without the bracket and with fixed α it is plain Coulomb, the test case of Sec. 5.1.) → `V_pert`

Joining to the long-distance potential, [JKT] (15). (13) is valid only at large Q; at small Q the coupling blows up at $Q = \Lambda$. Below $|\vec p| = 5$ GeV JKT switch to a Richardson-like form that also produces the linear confining potential:

$$\tilde V(\vec p) = \begin{cases} \tilde V_{\rm pert}(p^2, n_F = 5), & |\vec p| > 5\ \text{GeV}, \\ -\dfrac{4}{3}\,\dfrac{48\pi^2}{27}\,\dfrac{1}{p^2}\left(\dfrac{1}{\ln(1 + p^2/\Lambda_R^2)} + C\right), & |\vec p| < 5\ \text{GeV}, \end{cases} \qquad \Lambda_R = 400\ \text{MeV}.$$

C is fixed by continuity at 5 GeV: setting the two lines equal at $p^2 = 25\ \text{GeV}^2$ and solving for C (one linear equation) gives $C = -0.0262$. At small p, $1/\ln(1 + p^2/\Lambda_R^2) \approx \Lambda_R^2/p^2$, so $\tilde V \approx -\kappa\Lambda_R^2/p^4$ with $\kappa = \frac{4}{3}\cdot\frac{48\pi^2}{27}$: the Fourier transform of a potential rising linearly with r. → `V_ir, continuity_constant_C, V_jkt`

The cut, [JKT] (19). The $1/p^4$ behaviour is a problem: its Fourier integral diverges at $p \to 0$, and the kernel (18) built from it is singular as $(p-q)^{-2}$ at $q = p$. JKT argue that t and $\bar t$ decay before they reach the confining distances, so G cannot depend on how the potential behaves at very small momentum, and they replace it there:

$$\tilde V_{\rm cut}(\vec p; q_{\rm cut}) = \begin{cases} \tilde V(\vec p), & |\vec p| > q_{\rm cut}, \\ -\dfrac{C_1}{p^2 + q_{\rm cut}^2} + \tilde V_0(q_{\rm cut})\,\delta^3(\vec p), & |\vec p| < q_{\rm cut}, \end{cases} \qquad q_{\rm cut} = 50\ \text{MeV}.$$

C_1 comes from continuity at $p = q_{\rm cut}$: $-C_1/(2q_{\rm cut}^2) = \tilde V(q_{\rm cut}^2)$.

Normalising the cut, [JKT] (20). Cutting changes $V(r)$ by a constant. JKT fix $\tilde V_0$ by demanding that the position-space potential

$$V_{\rm cut}(r; q_{\rm cut}) = \int \frac{d^3q}{(2\pi)^3}\,e^{-i\vec q\cdot\vec r}\,\tilde V_{\rm cut}(\vec q; q_{\rm cut}) = \frac{1}{2\pi^2 r}\int_0^\infty dq\, q\sin(qr)\,\tilde V_{\rm cut}(q^2) + \frac{\tilde V_0}{(2\pi)^3}$$

satisfy $V_{\rm cut}(r = 1\,\text{GeV}^{-1}) = -\frac{1}{4}$ GeV. The δ^3 term is just the constant $\tilde V_0/(2\pi)^3$ in $V(r)$, i.e. a constant added to H, so in the Green's function it is the same as $E \to E - \tilde V_0/(2\pi)^3$. I therefore solve with the smooth part of (19) and evaluate G_0 at the shifted energy. For $\alpha_s(m_Z) = 0.12$ the shift is $+5.678$ GeV. It is large because it cancels the constant $\sim \kappa\Lambda_R^2/(2\pi^2 q_{\rm cut})$ that the cut adds to $V(r)$; it depends on $q_{\rm cut}$, but G does not (Sec. 5). → `V_cut, V_cut_position, energy_shift`

4 Solving the Lippmann–Schwinger equation (JKT Sec. 3)

Step 1: the equation, [JKT] (5), (16)–(17). The Green's function obeys

$$G(\vec{p}) = G_0(p) + G_0(p) \int \frac{d^3q}{(2\pi)^3}\, \tilde{V}(\vec{p}-\vec{q})\, G(\vec{q}) \qquad ([\text{JKT}]\ (5)).$$

For the S-wave JKT write $G = G_0 K$ (16), and the vertex function K obeys

$$K(p) = 1 + \int_0^\infty dq\, \hat{V}(p,q)\, G_0(q)\, K(q) \qquad ([\text{JKT}]\ (17)).$$

K is smooth in p: the sharp structure of G (the on-shell peak at $p^2 = m_t E$ for $E > 0$) sits in G_0, which is known exactly.

Step 2: the S-wave kernel, [JKT] (18). The angular integral of $\tilde{V}(|\vec{p}-\vec{q}|)$ gives

$$\hat{V}(p,q) = \frac{1}{4\pi^2}\frac{q}{p}\int_{|p-q|}^{p+q} du\, u\, \tilde{V}(u) = \frac{q}{8\pi^2 p}\Big[F\big((p+q)^2\big) - F\big((p-q)^2\big)\Big],$$
$$F(s) = \int_0^s dk^2\, \tilde{V}(k^2).$$

F is computed once per potential (10-point Gauss–Legendre on 6000 cells in $\ln s$, with the kinks of $\tilde{V}$ at q_{cut} and 5 GeV on cell edges) and splined. This is the only place the potential of Sec. 3 enters the solver. For pure Coulomb, $\tilde{V} = -B/k^2$, the integral is elementary: $\hat{V} = -\frac{Bq}{8\pi^2 p}\ln\frac{(p+q)^2}{(p-q)^2}$.
→ `LSSolver.kernel`

Step 3: the peak at $q = p$, and JKT's subtraction, [JKT] (21)–(22). $\hat{V}(p,q)$ is large at $q = p$ (small momentum transfer). With the cut it is finite but peaked on the scale $q_{\text{cut}} = 50$ MeV; for Coulomb it has a logarithmic singularity. The grid nodes (Step 5) are ~ 0.1–0.25 GeV apart, so a plain Gauss–Legendre sum misses the peak (Fig. 1, left).

Why not simply put more nodes there? Because "there" is everywhere: the same nodes q_j serve every row $p = q_i$ of the linear system (Step 6), and the peak of row i sits at $q = q_i$, on a node of its own. Refining the grid around one p does nothing for the others; resolving a 50 MeV peak at every node would take a grid ∼100 times finer, $N \sim 10^4$. JKT instead add and subtract $K(p)$ inside the integral of (17):

$$\big[1 - \mathcal{A}(p)\big]K(p) + \int_0^\infty dq\, \hat{V}(p,q)\, G_0(q)\, \big[K(p) - K(q)\big] = 1, \qquad ([\text{JKT}]\ (21))$$
$$\mathcal{A}(p) = \int_0^\infty dq\, \hat{V}(p,q)\, G_0(q). \qquad ([\text{JKT}]\ (22))$$

This is an identity, not an approximation. It moves the peak into $\mathcal{A}(p)$, which contains no unknown: in JKT's words, "a known function which can be evaluated by a numerical integration". In the remaining integral the peak is multiplied by $K(p) - K(q) \approx K'(p)(p-q)$, which vanishes at $q = p$. What is left is ~ 1000 times smaller and odd about $q = p$, so the contributions of the nodes on either side largely cancel (Fig. 1, right). The peak is then integrated *once per* p, in $\mathcal{A}(p)$, where a fine rule built around that one p is cheap (Step 4, Fig. 1, middle).

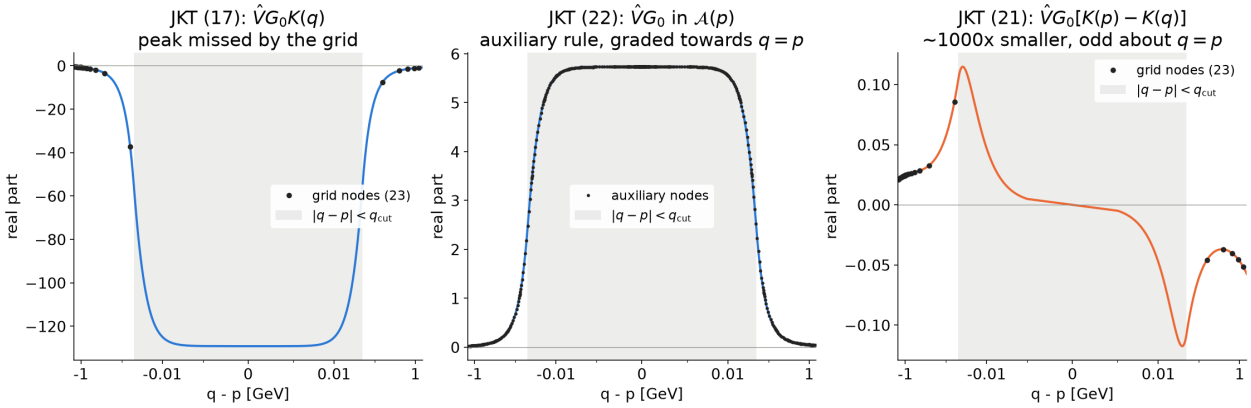


Figure 1: JKT potential ($\alpha_s(m_Z) = 0.12$, $q_{\text{cut}} = 50$ MeV), $p = 5$ GeV, $E = -2$ GeV, against $q - p$ on a logarithmic scale (linear for $|q - p| < 1$ MeV) so that the peak region, $|q - p| \lesssim q_{\text{cut}}$ (shaded), is spread out. Left: the integrand of [JKT] (17) with the grid nodes of [JKT] (23): the peak lies between two nodes. Middle: the integrand of [JKT] (22), on the auxiliary rule of $\mathcal{A}(p)$, whose panels are graded geometrically towards $q = p$ and so are evenly spaced on this axis. Right: the integrand of [JKT] (21), $\sim 1000\times$ smaller and odd about $q = p$, which the grid nodes integrate accurately.

Step 4: computing $\mathcal{A}(p)$, [JKT] (22). JKT do not say how. I use a fine auxiliary Gauss–Legendre rule for each p: 16-point panels with edges graded geometrically towards $q = p$ ($p(1 \pm 2^{-k})$, $k \leq 24$, i.e. evenly spaced in $\ln|q - p|$; Fig. 1, middle) to resolve the peak, further edges at $p(1 + 2^k)$ for its fall-off, every 2 GeV up to 200 GeV for the on-shell peak of G_0, and a $t = 1/q$ tail. $\hat{V}$ on these points is stored once, so for each energy $\mathcal{A}(p)$ is one dot product with G_0. Against adaptive quadrature the relative error is 2×10^{-10} (Coulomb) and 6×10^{-6} (JKT, limited by the spline of F). → `LSSolver.aux_rule, script_A`

Step 5: the grid, [JKT] (23). Every $\int_0^\infty dq$ in (21) becomes a Gauss–Legendre sum $\sum_j w_j(\cdot)(q_j)$. JKT use three subintervals, $(0, a_1)$, (a_1, a_2) and $(0, 1/a_2)$ in $t = 1/q$, with $a_1 \sim \sqrt{m_t\Gamma_t}$ and $a_2 \sim m_t$. I split them once more, at 4 and 60 GeV: edges $0, 4, 16, 60, m_t$ plus the $t = 1/q$ tail, with $n = 48$ points each ($N = 240$; JKT used $N = 100$–400). The reason: above threshold at narrow width ($m_t = 120$, $\Gamma_t = 0.3$ GeV, JKT's Fig. 4a at $E = 2$), the on-shell peak of G_0 at $p_0 = \sqrt{m_t E}$ is only $m_t\Gamma_t/2p_0 \approx 1.2$ GeV wide. On JKT's wide middle panel it falls between the nodes, and the peak of $|pG|^2$ goes $9272 \to 7680 \to 7219$ for $n = 48, 96, 160$ without settling. With the extra breaks it gives $7065, 7044, 7036$ (JKT: ~ 7050). → `LSSolver.jkt_grid`

Step 6: one linear system per energy. Imposing (21) at the nodes $p = q_i$ gives, with $g_j = G_0(q_j)$ and $W_{ij} = w_j\hat{V}(q_i, q_j)$,

$$\Big[1 - \mathcal{A}(q_i) + \sum_{j\neq i} W_{ij}g_j\Big]K_i - \sum_{j\neq i} W_{ij}g_jK_j = 1.$$

The $j = i$ term drops out exactly, because it multiplies $K_i - K_i = 0$. W does not depend on E and is built once. Each energy costs $\mathcal{A}$ (Step 4) and one dense $N \times N$ solve. → `LSSolver.solve_K`

Step 7: G at any p, [JKT] (16). $G(p) = G_0(p)K(p)$, with K between the nodes from a cubic spline in $\ln q$ through the $K(q_i)$. Because K is smooth and G_0 is exact, this keeps the sharp structure

of G. (I also tried evaluating (21) itself at off-grid p. That is 10–40 times less accurate, because at a general p the nodes are not placed symmetrically around the peak of Step 3.) → `LSSolver.G`

5 Is the solver right?

5.1 Ground truth: the exact Coulomb solution

For a fixed-coupling Coulomb potential, $V = -C_F\alpha/r$, the equation can be solved on paper (JKT mention this below their (12)). The solution is a Whittaker function,

$$G(r) = -\frac{m_t}{4\pi r}\,\Gamma(1-\lambda)\,W_{\lambda,1/2}(2\kappa r), \qquad \kappa = \sqrt{-m_t\mathcal{E}}, \quad \lambda = \frac{C_F\alpha m_t}{2\kappa},$$

and its Laplace representation plus the S-wave Fourier transform give a one-dimensional integral,

$$\tilde{G}(p) = -4m_t\kappa^2\int_0^\infty dt\,\frac{t^{-\lambda}(1+t)^{\lambda}(1+2t)}{\big(\kappa^2(1+2t)^2+p^2\big)^2},$$

which is G_0 at $\lambda \to 0$. It converges at $t \to 0$ only for $\mathrm{Re}\,\lambda < 1$ ($\lambda = 1$ is the 1S pole), so it is continued by subtracting $f(0)$ on $[0,1]$, valid for $\mathrm{Re}\,\lambda < 2$; since $|\lambda| \le 1.08$ for $\alpha = 0.15$, one subtraction suffices. No numerical solver is involved. → `CoulombExact`

5.2 Accuracy and $q_{\rm cut}$ independence (Stage 0)

On $E \in [-6, 4]$ GeV, $p \in [0.5, 29.5]$ GeV, $m_t = 173$, $\Gamma_t = 1.49$ GeV. The solver with a Coulomb potential is compared with the exact solution; with the JKT potential, against itself at $n = 96$:

		Coulomb $\alpha = 0.15$ vs exact		JKT $\alpha_s(m_Z) = 0.12$ vs $n = 96$	
n	N	median	max	median	max
16	80	5.3×10^{-5}	1.1×10^{-2}	6.5×10^{-4}	8.9×10^{-2}
24	120	1.4×10^{-5}	2.2×10^{-3}	2.4×10^{-4}	2.1×10^{-2}
32	160	5.9×10^{-6}	7.0×10^{-4}	1.2×10^{-4}	8.9×10^{-3}
48	240	1.8×10^{-6}	2.0×10^{-4}	5.4×10^{-5}	3.2×10^{-3}
64	320	7.5×10^{-7}	8.0×10^{-5}	2.5×10^{-5}	1.4×10^{-3}

The maxima sit above threshold on the on-shell ridge; for $E \le 0$ they are 1.1×10^{-5} (Coulomb) and 5.8×10^{-4} (JKT) at $n = 48$. JKT quote "better than 1%, typically 10^{-3}".

$q_{\rm cut}$ *independence* (JKT: "K is independent from $q_{\rm cut}$ for $q_{\rm cut} < 200$ MeV"). Relative to $q_{\rm cut} = 50$ MeV, G changes by at most 4.9×10^{-3} at 200 MeV, 1.6×10^{-3} at 100 MeV, 1.0×10^{-3} at 20 MeV and 1.3×10^{-3} at 10 MeV, while the energy shift of (20) ranges from 1.6 to 27 GeV. The cut does what JKT claim: it regularises without changing the physics.

5.3 Reproducing JKT's own results (Stage 1)

Since this is JKT's method, it must reproduce their figures (Fig. 2). I use $\alpha_s = 0.125$ as JKT do, and Γ_t from [JKT] (8)–(9): 0.299, 0.807 and 1.563 GeV for $m_t = 120$, 150 and 180 (JKT's text rounds to 0.3/0.8/1.5). At $m_t = 120$ the 1S peak of R is at -2.32 GeV (JKT: -2.3) with height 3.80 (JKT: 3.78). The 1S–2S minimum, the 2S bump, $R(0)$ and $R(4)$ agree to 0.2–2%. The peaks of $|pG|^2$ in their Fig. 4a–c agree to 0–3% with the values read off by eye, and $-G(p \to 0)$ in their Fig. 2 to 0.4–3%. The exception is $\mathrm{Im}(-G)(p \to 0)$ at $E = 0$ (Fig. 3): -2.98 against their -2.7, i.e.

10%, on a small value on a ± 8 axis. The optical theorem [JKT] (12), $\int \frac{d^3p}{(2\pi)^3}\Gamma_t|G|^2 = -\mathrm{Im}\,G(0,0)$, holds to 4×10^{-5} for $E\in[-5,5]$ GeV; it is not built into the solver, so it independently checks the solution.

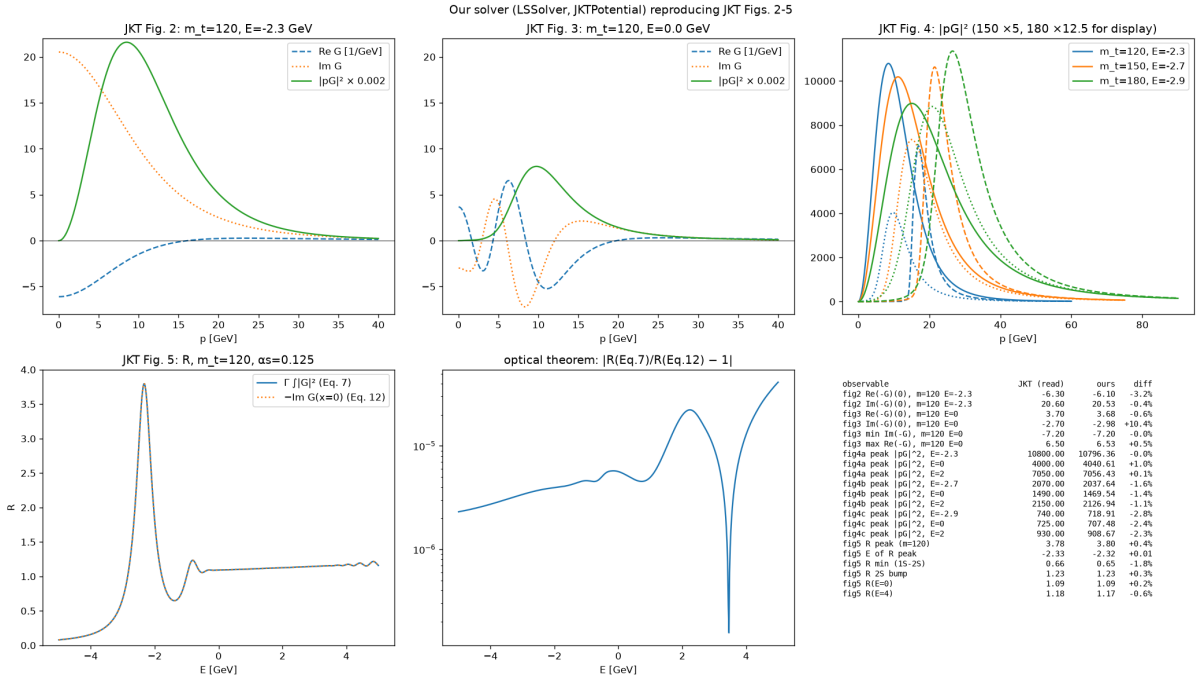


Figure 2: JKT's Figs. 2–5 recomputed with my solver ($\alpha_s = 0.125$, Γ_t from [JKT] (8)). Top: G and $|pG|^2$ at $m_t = 120$ GeV at the 1S peak and at threshold; $|pG|^2$ for $m_t = 120, 150, 180$ GeV (solid, dotted, dashed: below, at and above threshold). Bottom: the cross-section ratio $R(E)$ from $\int \Gamma_t|G|^2$ and from $-\mathrm{Im}\,G(0,0)$ (the optical theorem [JKT] (12)), their relative difference, and the comparison with values read off JKT's figures. (`1_validate_jkt.py`)

One convention to watch: JKT's Figs. 1–3 plot $-G$ relative to their (6) and (10). At the 1S peak their Fig. 2 has $\mathrm{Im} \approx +20$ GeV^{-1} at $p \to 0$, whereas $\mathrm{Im}\,G = -|\psi(0)|^2/\Gamma_t < 0$.

6 Fuks et al.'s tables and figures

What Fuks et al. do. They re-weight colour-singlet $gg \to t\bar{t} \to b\ell^+\nu\,\bar{b}\ell'^-\bar{\nu}$ MadGraph events by $|G/G_0|^2$ ([Fuks] (15)), with G interpolated from their swData tables by their Fortran routine `CALCGREEN`. They describe the potential behind the tables in four different ways (running tree-level coupling, Richardson following [JKT], fixed $\alpha_s(m_Z) = 0.12$, two-loop) and do not describe the method. I read their tables and run their routine exactly as published, as a reference to compare with. → `FuksTables`

What is in their tables (Stage 2). Fitting each of their descriptions to swData, using my solver and the exact Coulomb solution (rms of |swData/model − 1| on $E \in [-6, 4]$, $p < 30$ GeV):

Potential	rms
fixed-α tree Coulomb, best fit $\alpha = 0.1500$, $m_t = 172.96$, $\Gamma_t = 1.4907$	0.13%
fixed $\alpha = 0.12$ (their Fig. 1 caption), with or without the bracket of (13)	40–50%
JKT potential (13)–(15), $\alpha_s(m_Z) = 0.12$, tree or NLO	55–69%
the same, with a free energy offset	15–17%

So swData is the exact fixed-coupling Coulomb Green's function at $\alpha = 0.15$, not the JKT potential. A plausible origin: $\alpha_s(m_Z) = 0.12$ run with (14) reaches 0.150 at $Q \approx 25$ GeV, close to the Bohr momentum (Sec. 3). The authors state neither. My solver with Coulomb $\alpha = 0.15$ reproduces their tables as well as the exact solution does (median deviation 2.6×10^{-3} for both, Stage 3, Fig. 3), which is a further check of the solver.

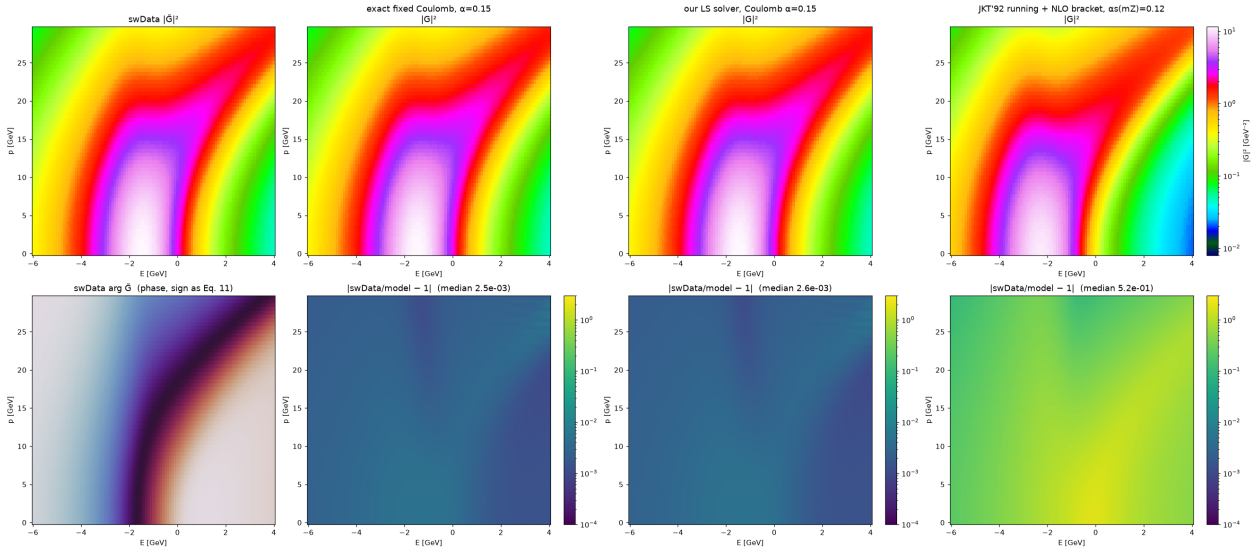


Figure 3: Fuks's Fig. 1, rebuilt from their swData (left column: $|\tilde{G}|^2$ and phase), next to exact Coulomb at $\alpha = 0.15$, my solver with Coulomb at $\alpha = 0.15$, and my solver with the JKT potential at $\alpha_s(m_Z) = 0.12$. Bottom row: |swData/model − 1|. (`3_fuks_fig1.py`)

Events (Stage 4). 13 TeV, CT18NLO, $m_t = 173$, $\Gamma_t = 1.49$: 500k colour-singlet ($330 \le W \le 350$ GeV, colour matrix [Fuks] (20)) and 500k full colour ($340 \le W \le 350$); the window is imposed by restricting $\tau = \hat{s}/s$ in MadGraph's `genps.f`. The singlet fraction is 0.281 (expected 2/7). The same events are re-weighted with three sources of G:

	Fuks: swData via `CALCGREEN`	Coulomb $\alpha = 0.15$ exact	**JKT potential (this note)** $\alpha_s(m_Z) = 0.12$
$\sigma_{\rm rw}/\sigma$ ($p^* < 50$ GeV)	8.88	8.90	7.89
$\langle p^* \rangle (E = -2)$ [GeV]	20.7	20.7	19.6
peak of $d\sigma/dW$ [GeV]	344.75	344.75	343.75

With Fuks's own G the re-weighted distributions reproduce their Figs. 2–5 (Figs. 4–7). The JKT potential, i.e. the physics they cite, gives an enhancement smaller by 11%, a peak 1 GeV lower and softer p^*. → `Events`

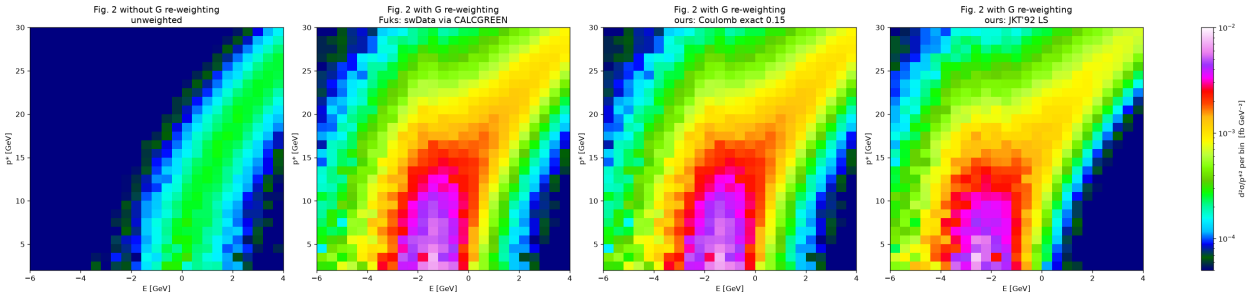


Figure 4: Fuks's Fig. 2 from our events: $d^2\sigma/p^{*2}$ per bin in the (E, p^*) plane, without re-weighting and re-weighted with Fuks's G, exact Coulomb at $\alpha = 0.15$, and the JKT potential. (`4_fuks_figs2to5.py`)

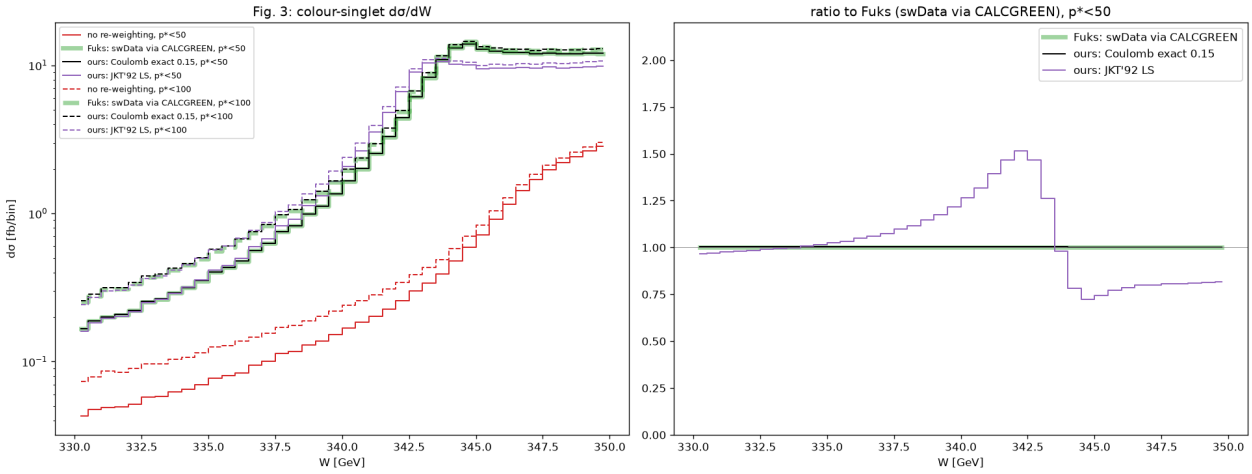


Figure 5: Fuks's Fig. 3: colour-singlet $d\sigma/dW$ in 0.5 GeV bins, without re-weighting and re-weighted with the three sources of G, for $p^* < 50$ and $p^* < 100$ GeV; right, the ratio to Fuks's own G. Exact Coulomb at $\alpha = 0.15$ lies on top of Fuks; the JKT potential moves the peak down by about 1 GeV.

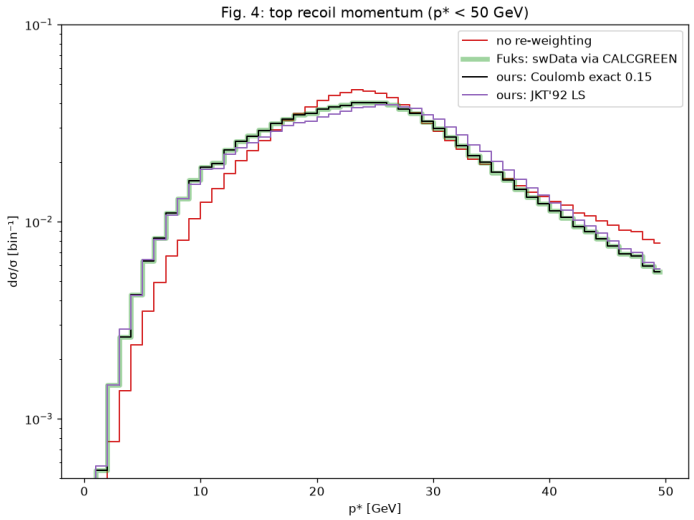


Figure 6: Fuks's Fig. 4: the distribution of the top recoil momentum p^* ($p^* < 50$ GeV, fraction per 1 GeV bin), without re-weighting and re-weighted with the three sources of G.

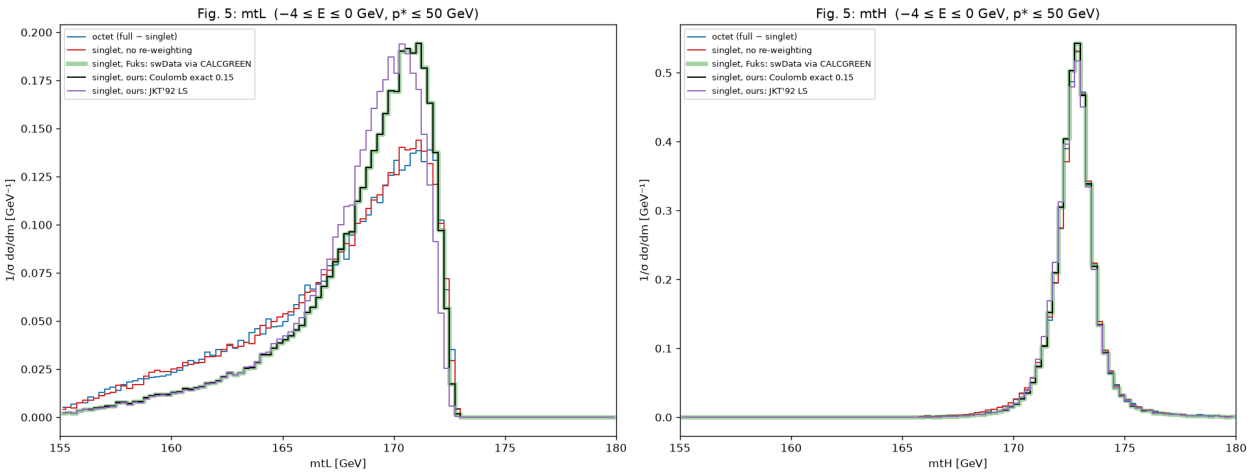


Figure 7: Fuks's Fig. 5: the lighter and heavier reconstructed top mass, m_{t_L} and m_{t_H}, for $-4 \leq E \leq 0$ GeV and $p^* \leq 50$ GeV: colour octet, singlet without re-weighting, and singlet re-weighted with the three sources of G.

7 Toponium in $pp \to t\bar{t}$ events: K-factors at 13 TeV

The Green's function of Secs. 3–4 is put into MadGraph's matrix element so that event samples contain toponium. The effect is quoted per bin of measured $t\bar{t}$ distributions as a *K-factor*,

$$K = \frac{\sigma_{\text{with threshold effects}}(\text{bin})}{\sigma_{\text{LO}}(\text{bin})} = 1 + \frac{\Delta}{\text{SM}_{\text{full}}},$$

with SM_{full} the LO cross section in the bin and Δ the change that the threshold effects make to it. The model has both colour states of the pair: the attractive singlet (toponium, from our G) and the repulsive octet (Sec. 7.4). → `6_tfactors.py`

7.1 Colour: singlet and octet in $pp \to t\bar{t}$

At LO, $pp \to t\bar{t}$ has two channels.

- $q\bar{q} \to t\bar{t}$: one diagram, $q\bar{q} \to g^* \to t\bar{t}$. A single gluon is a colour octet, so the pair it produces is an octet. In the amplitude the colour factor is $(T^a)_{ij}(T^a)_{kl}$; projecting $t_k\bar{t}_l$ on the singlet ($\delta_{kl}/\sqrt{3}$) gives $\text{Tr}\, T^a = 0$. $q\bar{q}$ gives singlet pairs only via γ^*/Z (not in our pure-QCD samples) or beyond LO.
- $gg \to t\bar{t}$: three diagrams, singlet *and* octet ($8 \otimes 8 \ni 1$). MadGraph writes the amplitude in two colour flows, $\mathcal{M} = J_1(T^aT^b)_{kl} + J_2(T^bT^a)_{kl}$, and $|\mathcal{M}|^2 = \sum_{ij} J_i^*\, C_{ij}\, J_j$ with

$$C = \tfrac{1}{3}\begin{pmatrix} 16 & -2 \\ -2 & 16 \end{pmatrix} = \underbrace{\tfrac{2}{3}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}}_{C_1\ \text{(singlet, [Fuks] (19)–(20))}} + \underbrace{\tfrac{1}{3}\begin{pmatrix} 14 & -4 \\ -4 & 14 \end{pmatrix}}_{C_8\ \text{(octet)}}, \qquad |\mathcal{M}|^2 = |\mathcal{M}_1|^2 + |\mathcal{M}_8|^2,$$

  with no singlet–octet interference. Near threshold the singlet is about 2/7 of gg.

The two colour states feel different potentials. The singlet's is attractive, $V_1 = -C_F\alpha_s/r$ at short distance, and is the one behind G (Secs. 3–4). The octet's is repulsive, $V_8 = +\alpha_s/(6r)$, with colour factor $C_F - C_A/2 = -1/6$ [HSY]: it has no bound states and suppresses the rate near threshold. About 84% of the threshold rate in $pp \to t\bar{t}$ is octet, so both are included. Toponium itself forms only in the gg colour singlet, in the S-wave that our G describes.

7.2 The matrix-element modification

Per event, $E = W - 2m_t$ with $W = m_{t\bar{t}}$, and p^* is the top momentum in the $t\bar{t}$ rest frame ([Fuks] (21)). Inside the threshold window ($W \leq 350$ GeV, p^* below a cut, and $E \geq -20$ GeV, the range of our tables) each colour state's matrix element is multiplied by its own $|G/G_0|^2$; outside it nothing changes:

$$gg:\ |\mathcal{M}|^2 \to \left|\frac{G_1}{G_0}\right|^2 |\mathcal{M}_1|^2 + \left|\frac{G_8}{G_0}\right|^2 |\mathcal{M}_8|^2, \qquad q\bar{q}:\ |\mathcal{M}|^2 \to \left|\frac{G_8}{G_0}\right|^2 |\mathcal{M}|^2.$$

This is [Fuks] (15) applied to each colour state. The singlet part is their (16), $|\mathcal{M}|^2 + (|G_1/G_0|^2 - 1)|\mathcal{M}_1|^2$, and the octet is treated in the same way. $|G/G_0|^2$ is read in the Fortran from tables of our G_1 and G_8 (bilinear interpolation, a module modelled on [Fuks]'s `GREENDATA/CALCGREEN`), and $|\mathcal{M}_1|^2$ is computed alongside $|\mathcal{M}|^2$ with C_1, with $|\mathcal{M}_8|^2 = |\mathcal{M}|^2 - |\mathcal{M}_1|^2$. The modified matrix element stays positive, so MadGraph generates it directly; no reweighting is done afterwards. → `F_GREEN_FACTOR, patch_matrix1, patch_qq`

7.3 α_s: CTEQ's running versus [JKT] (14)

0.118 is CT25NNLO's $\alpha_s(m_Z)$; its $\alpha_s(Q)$ runs, but differently from [JKT] (14): CTEQ solve the renormalisation-group equation numerically at three loops with a variable number of flavours (thresholds at m_c, m_b) and tabulate it; (14) is a two-loop closed form with fixed $n_F = 5$. Both agree at m_Z by construction:

Q [GeV]	5	10	25	91.19
CT25NNLO (LHAPDF)	0.21257	0.17815	0.14688	0.11800
[JKT] (14), two-loop, $\alpha_s(m_Z) = 0.118$	0.21396	0.17864	0.14705	0.11800

MadGraph takes $\alpha_s(Q)$ for the matrix elements from the PDF. For one α_s everywhere, the potentials behind G_1 and G_8 use CT25's $\alpha_s(Q)$ too, read from LHAPDF (for the singlet it enters only above 5 GeV and through C). This changes C from -0.0320 to -0.0332 and $|G_1/G_0|^2$ by at most 0.1%. Formally it combines three-loop running with the one-loop bracket of (13); the effect is far below the other approximations. → `JKTPotential(alpha_fn=...)`

7.4 The octet potential

For the octet, (13) with the colour factor $C_F \to C_F - C_A/2 = -1/6$ [HSY]:

$$\tilde{V}_8(q) = +\frac{1}{6}\,\frac{4\pi\alpha_s(Q^2)}{q^2}\left[1 + \left(\frac{31}{3} - \frac{10}{9}n_F\right)\frac{\alpha_s(Q^2)}{4\pi}\right], \qquad Q = \max(q, Q_0),$$

with α_s CT25's. The potential is repulsive, so the long-distance problem the singlet has does not arise: there is no bound state to build, and G_8 is insensitive to the confining region. It is enough to stop the running before α_s grows large, by freezing it below a scale Q_0. Below Q_0 the potential is then exactly Coulomb, so it needs neither the cut (19) nor an energy shift, and G_8 is solved with the same `LSSolver` as the singlet, with Q_0 as an extra breakpoint of F. [JKT] have no octet; this form is my choice (Sec. 8). Four versions span the uncertainty: $Q_0 = 10$, 25, 50 GeV ("freeze10/25/50") and fixed-coupling tree-level Coulomb at $\alpha_s = \alpha_s^{\mathrm{CT25}}(25\ \mathrm{GeV}) = 0.14688$ ("coulomb_as25", exact solution of Sec. 5.1).

$\lvert G_8/G_0\rvert^2$ at (E,p) [GeV]	freeze10	freeze25	freeze50	coulomb_as25
$(-2,\ 5)$	0.6765	0.6977	0.7179	0.6906
$(0,\ 10)$	0.6535	0.6744	0.6964	0.6679
$(+4,\ 30)$	0.7326	0.7510	0.7660	0.7386
$(-2,\ 50)$	0.8329	0.8384	0.8451	0.8227
$(+2,\ 75)$	0.8642	0.8674	0.8714	0.8494

Checks: `octet_frozen` at constant α_s against the exact Coulomb solution, 2.8×10^{-7}; the Fortran tables against the solver, median $\sim 3 \times 10^{-8}$, max 1.6×10^{-5}; doubling the grid ($n = 120 \to 160$), below 10^{-8}. The octet lowers the rate in the window by 13–35%, so overall it removes 13.5% of the singlet's toponium excess (Sec. 7.7; 15% in the earlier full-sample runs, Sec. 7.11). → `JKTPotential.octet_frozen; table_window`

7.5 From events to K-factors

Outside the threshold window the modified and the LO matrix elements are identical, so Δ comes only from events inside the window. The samples are therefore split:

1. $\mathrm{SM}_{\mathrm{full}}$: the LO cross section per bin from one full sample, $pp \to t\bar{t} \to W^+ b\, W^- \bar{b}$, $W \to \ell\nu$ ($\ell = e, \mu$, all four combinations, [Fuks]'s decay-chain syntax), 10×10^6 events (seeds 7101–7191), no window, $\sigma = 20.1317$ pb.
2. Δ: samples generated only in the threshold region ($W \leq 350$ GeV through `TAUMAX` in `genps.f`, $p^* \leq 100$ GeV through `dummy_cuts`): $\mathrm{SM}^{gg}_{\mathrm{win}}$, $\mathrm{SM}^{q\bar{q}}_{\mathrm{win}}$ with the LO matrix element and $\mathrm{topo}^{gg}_{\mathrm{win}}$, $\mathrm{topo}^{q\bar{q}}_{\mathrm{win}}$ with the modified one ($|G/G_0|^2$ applied for $p^* < 80$ GeV), 10^6 events each. The $q\bar{q}$ samples contain both beam orderings, $q\bar{q}$ and $\bar{q}q$ (`generate q q~ > ...` plus `add process q~ q > ...`; Sec. 7.8). Then per bin

$$\Delta = (\mathrm{topo}^{gg}_{\mathrm{win}} - \mathrm{SM}^{gg}_{\mathrm{win}}) + (\mathrm{topo}^{q\bar{q}}_{\mathrm{win}} - \mathrm{SM}^{q\bar{q}}_{\mathrm{win}}) \quad \text{restricted to } 326 \leq W \leq 350 \text{ GeV},\ p^* < p_{\mathrm{cut}},$$

   with $p_{\mathrm{cut}} = 100$ or 50 GeV. Each $\mathrm{topo}_{\mathrm{win}}$ sample is generated once per octet potential.
3. *Reconstruct.* Top $= b + \ell^+ + \nu$, antitop $= \bar{b} + \ell^- + \bar{\nu}$, from the final-state partons (off-shell tops). Observables: $m_{t\bar{t}}$, $y_{t\bar{t}}$, $|y_{t\bar{t}}|$, $|y_t + y_{\bar{t}}|/2$, $H_T = p_T^t + p_T^{\bar{t}}$, p_T^t, $|y_t|$ and the 2D $|y_t| \times p_T^t$, in all the requested binnings (12). Every event carries weight $\sigma_{\mathrm{run}}/N_{\mathrm{run}}$.
4. *Errors* (MC statistics): $\delta\Delta$ from the four window samples in quadrature, and

$$\delta K = \sqrt{(\delta\Delta/\mathrm{SM}_{\mathrm{full}})^2 + (\Delta\, \delta\mathrm{SM}_{\mathrm{full}}/\mathrm{SM}^2_{\mathrm{full}})^2}.$$

Generating only in the window puts all the statistics where the effect is: in the lowest $m_{t\bar{t}}$ bin $\delta K = 9 \times 10^{-5}$, against 1×10^{-3} from two independent 10^7-event full samples (Sec. 7.11). This design is the student's (Sec. 7.9). All the samples are run 6, `run6_final_kfactors_13TeV`: 2×10^7 events for the K-factors and 3×10^6 more for the checks of Sec. 7.8. → `kfactors`

7.6 Results

The final K-factors are in `results/6_tfactors/run6_final_kfactors_13TeV/kfactors_final/`, one plain CSV per binning (bin edges, K, its MC error, a model uncertainty, and K_{norm} for distributions normalised to their own integral). The model uncertainty adds in quadrature the change from $p^* < 100$ to $p^* < 50$ and the spread over the four octet potentials. They are drawn from the two full files one level up: the headline (`..._pstar100_octet-freeze25_ours.csv`) and the alternative with $p^* < 50$ and octet coulomb_as25 (`..._pstar50_octet-coulomb_as25_ours.csv`).

	$p^* < 100$, octet freeze25	$p^* < 50$, octet coulomb_as25
Δ (all bins) [pb]	0.16890	0.15790
total $K-1$	+0.839%	+0.784%
$m_{t\bar{t}} \in [250, 400]$ GeV	1.04212 ± 0.00009	1.03938 ± 0.00008
$m_{t\bar{t}} \in [345, 400]$ GeV	1.01994 ± 0.00006	1.01867 ± 0.00006
$H_T \in [0, 90]$ GeV	1.06360 ± 0.00013	1.06175 ± 0.00013
$H_T \in [90, 170]$ GeV	1.00158	1.00030
$p_T^t \in [0, 60]$ GeV	1.03935 ± 0.00008	1.03717 ± 0.00008
$\lvert y_{t\bar{t}} \rvert$ bins, 0 to 2.5	1.0078 rising to 1.0099	1.0074 rising to 1.0092
spread over the four octets	≤ 0.0006	≤ 0.0011

The effect sits where the tops are slow: the lowest $m_{t\bar{t}}$ bin (which contains the whole window), the lowest p_T^t and the lowest H_T bin; above them $K = 1$ exactly, since nothing changes there. In rapidity it is spread over all bins, rising towards large $|y|$ because the low-mass pairs reach larger rapidities. The $m_{t\bar{t}}$ binning that starts at 345 GeV shows about half the effect of the ones starting at 250 or 325 GeV: most of the toponium peak ($W \approx 343$–345 GeV) lies below its first edge. Going from $p^* < 50$ to $p^* < 100$ adds 7% to Δ. The choice of octet potential moves every K by about 10^{-3} at most (at most 4% of $K-1$), which is the uncertainty from the octet. Toponium is a gg effect: in the generation window the singlet raises the gg rate by a factor 2.45, while the octet lowers the $q\bar{q}$ rate by 21%. Figures 8 and 9 show the $p^* < 100$ file.

7.7 Singlet only versus singlet + octet

[Fuks]'s signal sample is $gg \to t\bar{t}$ in the colour singlet only, re-weighted by $|G_1/G_0|^2$: the octet gg pairs and all $q\bar{q}$ pairs are absent, and the octet gets no Green's function. To separate the two effects (the missing pairs, and the missing octet repulsion) in run 6's design, one extra window sample and two LO samples were generated:

1. $\text{topo}^{gg}_{\text{win}}$ with the octet table set to 1 (`topo_win_gg_none_s7850`): the same compiled MATRIX1 as run 6, so only the singlet is modified, [Fuks] (16). For $q\bar{q}$ (pure octet) singlet only means unmodified, i.e. $\text{SM}^{q\bar{q}}_{\text{win}}$.
2. $gg \to t\bar{t}$ at LO over all phase space, all colours (`SM_gg_full_s7860`) and the colour-singlet projection, [Fuks] (19)–(20) (`SM_gg1_full_s7870`, [Fuks]'s sample), 10^6 events each: the denominators for gg alone.

In the window ($326 \leq W \leq 350$ GeV, $p^* < 100$ GeV):

Δ, singlet only	$+0.19517 \pm 0.00034$ pb
Δ, singlet + octet (run 6)	$+0.16890 \pm 0.00032$ pb: gg +0.17705, $q\bar{q}$ −0.00815
octet effect	-0.02628 ± 0.00043 pb
as a fraction of the singlet-only excess	−13.5% (−13.4% at $p^* < 50$)

The same Δ divided by different LO samples gives the step from [Fuks]'s set-up to the full one ($K = 1 + \Delta/\text{LO}$ in the first bin, $p^* < 100$; MC errors ≤ 0.0013):

		LO [pb]	K		
sample	G applied to	$m_{t\bar{t}} \in [250, 400]$	$m_{t\bar{t}} \in [250, 400]$	$m_{t\bar{t}} \in [325, 400]$	$H_T \in [0, 90]$
gg, singlet events only ([Fuks])	singlet	0.852	1.2291	1.2294	1.3917
gg, all colours	singlet	3.201	1.0610	1.0611	1.0876
gg, all colours	singlet + octet	3.201	1.0553	1.0554	1.0795
pp	singlet	4.010	1.0487	1.0487	1.0735
pp (final, run 6)	singlet + octet	4.010	**1.0421**	**1.0422**	**1.0636**



Figure 8: K-factors per bin for all requested binnings ($p^* < 100$, octet freeze25): $K = 1 + \Delta/\mathrm{SM}_{\mathrm{full}}$ with its MC error; dark ticks, the same K with the other three octet potentials (visible only where they differ by more than the line width). Bins with no toponium effect have $K = 1$ exactly. The $p^* < 50$ figures are in `plots/pstar50/`. (`6_tfactors.py plotk`)

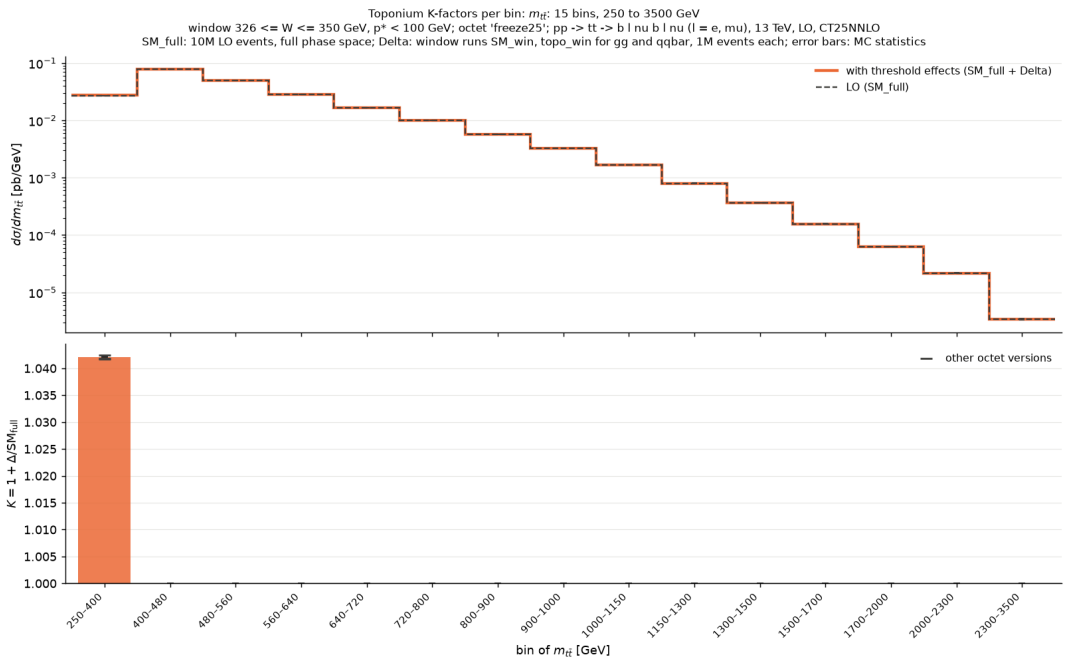


Figure 9: One binning in full ($m_{t\bar{t}}$, 250–3500 GeV): top, $d\sigma/dm_{t\bar{t}}$ at LO ($\mathrm{SM}_{\mathrm{full}}$, dashed) and with the threshold effects ($\mathrm{SM}_{\mathrm{full}} + \Delta$); bottom, K per bin. Every binning has such a figure in `plots/pstar100/` and `plots/pstar50/`.

Most of the step from [Fuks]'s $\approx 20\%$ (run 2: 1.209 at $p^* < 50$) to 4.2% is the denominator: the octet gg pairs and the $q\bar{q}$ pairs, about 4/5 of the pairs at threshold, join the LO rate while the singlet excess stays the same. The octet repulsion then lowers Δ itself, by $\Delta K = -0.0066 \pm 0.0001$ in the first $m_{t\bar{t}}$ bin (-0.0099 in the first H_T bin), in line with runs 1/3 (-0.0070 ± 0.0010, Sec. 7.11). The singlet-only row on top of pp (1.0487) is what [Fuks]'s signal gives when added to a full pp prediction. → `6_tfactors.py octetcheck run6_final_kfactors_13TeV`

7.8 Check of the window design: full phase space

Run 6 also generates the same physics over all phase space, so that every window sample has a full-phase-space counterpart: the LO sample $\mathrm{SM}_{\mathrm{full}}$ itself and $\mathrm{topo}_{\mathrm{full}}$ (pp, modified matrix element, freeze25, 10^6 events). In the window ($326 \le W \le 350$, $p^* < 100$) they must agree with $\mathrm{SM}_{\mathrm{win}}$ and $\mathrm{topo}_{\mathrm{win}}$. A first pass did not: $\mathrm{SM}_{\mathrm{win}}$ was 10% below $\mathrm{SM}_{\mathrm{full}}$ (25σ), and Δ from the full samples was 4–5% smaller, the same tension seen earlier between run 4 (full samples) and run 5 (window design, with the student's samples). Two further LO samples located the cause:

1. $\mathrm{SM}_{\mathrm{cut}}$: the window imposed by `dummy_cuts` alone, `genps.f` untouched. It equals $\mathrm{SM}_{\mathrm{win}}$ in every W and p^* bin (ratio 1.000 ± 0.002), so the `TAUMAX` window is not the cause.
2. $\mathrm{SM}^{500}_{\mathrm{cut}}$: $\mathrm{SM}_{\mathrm{full}}$'s own pp process and cards with only $W \le 500$ GeV imposed. It equals $\mathrm{SM}_{\mathrm{full}}$ (+3%), not $\mathrm{SM}_{\mathrm{win}}$.

So the difference was the process, not the window: `generate p p >`... includes the beam-swapped mirror $\bar{q}q$ of every $q\bar{q}$ subprocess, while `generate q q~ >`... (with `q = u c d s`) generates the quark only from beam 1 (`MIRRORPROCS = .FALSE.`). The window $q\bar{q}$ samples, ours and the student's, therefore held half of $q\bar{q}$. Since $q\bar{q}$ is pure octet and suppressed, this made Δ 2.4% too large ($K = 1.0431$ instead of 1.0421 in the lowest $m_{t\bar{t}}$ bin). With both orderings in the $q\bar{q}$ samples (final

run 6):

in $326 \le W \le 350$, $p^* < 100$	window samples	full phase space	pull
SM [fb]	$\mathrm{SM}_{\mathrm{win}}$ 153.03 ± 0.12	$\mathrm{SM}_{\mathrm{cut}}$ 153.08 ± 0.17, $\mathrm{SM}_{\mathrm{cut}}^{500}$ 153.4 ± 1.3	$+0.3$, $+0.3$
Δ [fb]	168.90 ± 0.32	$\mathrm{topo}_{\mathrm{full}} - \mathrm{SM}_{\mathrm{full}}$: 165.3 ± 2.6	-1.4
K, $m_{t\bar{t}} \in [250, 400]$	1.04212 ± 0.00008	1.0412 ± 0.0006	-1.4
K, $H_T \in [0, 90]$	1.06360 ± 0.00012	1.0622 ± 0.0010	-1.5
K, $p_T^t \in [0, 60]$	1.03935 ± 0.00007	1.0384 ± 0.0006	-1.5

One small residual remains: the two pp samples are 25–30% low against the window samples at $W = 326$–338 GeV, far below threshold (about 10% of the SM rate in the window). There a full-phase-space sample has very few events, while the window samples, made two different ways, agree with each other. It leaves $\mathrm{SM}_{\mathrm{full}}$ 2.7% low in the window as a whole, which does not enter K: Δ comes from the window samples, and the window is a small part of each bin. → `kfactors; arbiter`

7.9 Independent cross-check: the student's calculation

A graduate student in the group, Marcel Yáñez Reyes, computed the same K-factors independently, with their own Green's functions and their own MATRIX1 patch, in the design of Sec. 7.5 (their two CSV files, `kfactors_13TeV_W326-350_*.csv`, for $p^* < 100$ with freeze25 and $p^* < 50$ with coulomb_as25). Their singlet uses the same potential as ours (JKT with CT25 α_s; same C and energy shift), and their octets are the same four. Run 5 repeats the design exactly, with the same seeds, C definition and bins, but with our G_1, G_8 and our patch, so that any difference comes from the Green's functions or the patch.

	$p^* < 100$, freeze25	$p^* < 50$, coulomb_as25
Δ: ours / student [pb]	0.17307 / 0.17361 (-0.3%)	0.16199 / 0.16253 (-0.3%)
K column: bins with an effect	60	53
pull mean, rms	-0.32, 1.28	-0.42, 1.04
largest |pull|	4.3 ($y_{t\bar{t}}$ bin 1)	2.3
other octets: max $\lvert\Delta K\rvert$	$\le 4.9 \times 10^{-4}$	$\le 4.4 \times 10^{-4}$

Per sample, the cross sections agree to 0.04% or better for all five $q\bar{q}$ samples. The four gg samples with the Green's function are 0.09–0.18% below the student's. S_{gg} has no Green's function and is 0.10% below as well, so the small gg offset comes from generating the events, not from G. It accounts for the slightly negative pull mean. Apart from one $y_{t\bar{t}}$ bin at 4.3σ (1.00900 vs 1.00941, a difference of 4×10^{-4} in K), every bin agrees within 2.5σ. Two independent solvers and patches therefore give the same K-factors to a few 10^{-4}. The comparison was made with run 5, whose $q\bar{q}$ window samples, like the student's, lack the $\bar{q}q$ mirror (Sec. 7.8); it tests the Green's functions and the patch, which the missing half does not touch. The student's files should be regenerated with `add process q~ q > ...`; their $K - 1$ is about 2.4% too high. → `6_tfactors.py kfactors run5_window_kfactors_13TeV STUDENT_CSV ...`

7.10 Top-mass scan (run 7)

Run 6's design repeated at $m_t = 170$, 171.25, 172.5, 173.75 and 175 GeV (`run7_mt170p00_13TeV`, ...), run on the MSU HPCC (`submit_mtscan.sh`, `slurm_*.sh`). At each mass everything that depends on m_t follows it:

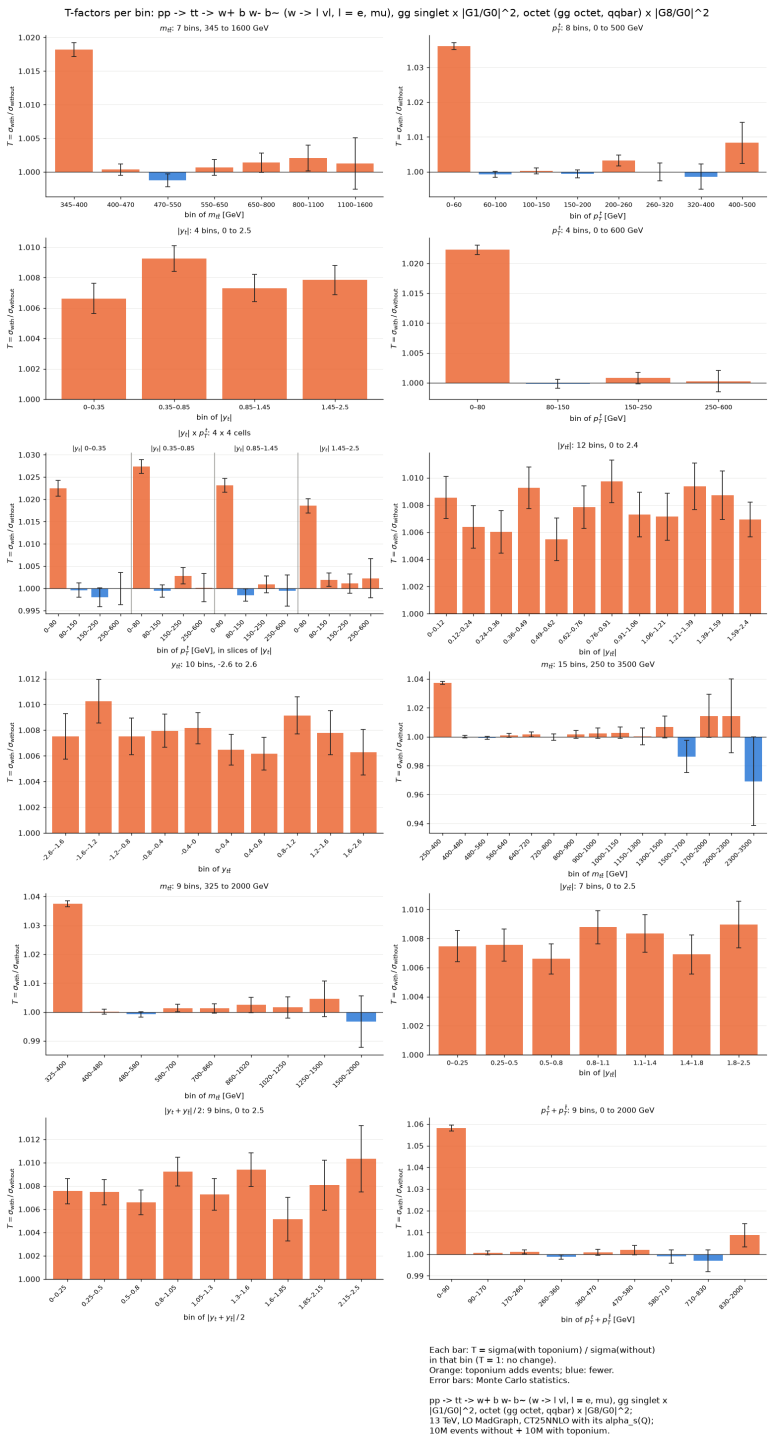


Figure 10: Run 4 ($pp \to t\bar{t}$, all e/μ combinations, singlet $\times|G_1/G_0|^2$ and octet $\times|G_8/G_0|^2$): T per bin for all requested binnings, from two independent 10^7-event full samples (the 2D $|y_t| \times p_T^t$ binning in slices of $|y_t|$). Error bars are MC statistics. Full samples include the $\bar{q}q$ mirror, so run 4 is a valid (if 10 times less precise) check: 1.0375 ± 0.0010 in the lowest $m_{t\bar{t}}$ bin at $p^* < 50$ against 1.0394 from the window design.

1. Γ_t from the LO $t \to Wb$ width, $\Gamma \propto m_t^3(1-x)^2(1+2x)$ with $x = m_W^2/m_t^2$, normalised to 1.49 GeV at 173 GeV (`top_width`): 1.4011, 1.4378, 1.4750, 1.5127, 1.5509 GeV. MadGraph's param card gets this m_t and Γ_t.
2. The free Green's function is that mass's: $G_0(E, p; m_t, \Gamma_t)$, [JKT] (6). `table_mt` solves the Lippmann–Schwinger equation (`LSSolver`) at that m_t and Γ_t and divides by $G_0(E, p; m_t, \Gamma_t)$; the octet table likewise (`octet_ratio`). Check: the values logged at three (E, p) points per mass, recomputed with each mass's G_0, agree within 7×10^{-5} (table interpolation); with $G_0(173, 1.49)$ they would differ by 2–10%. The 170 GeV table agrees with a direct solve to 10^{-11}.
3. The window moves with threshold: $E = W - 2m_t$ in the MATRIX1 patch, factors for $W \leq 2m_t + 4$, selection $2m_t - 20 \leq W \leq 2m_t + 4$ (326–350 GeV at 173 GeV).

Differences from run 6: the singlet uses [JKT]'s own two-loop α_s (14) at $\alpha_s(m_Z) = 0.12$ (not CT25NNLO's $\alpha_s(Q)$), and only one octet (coulomb_as25). Each mass has its own $\mathrm{SM}_{\mathrm{full}}$ (10×10^6 events) and window samples.

m_t [GeV]	170	171.25	172.5	173.75	175
K, $m_{t\bar{t}} \in [250, 400]$ (CMS)	1.0384	1.0404	1.0427	1.0452	1.0480
K, $m_{t\bar{t}} \in [325, 400]$ (ATLAS)	1.0382	1.0404	1.0428	1.0452	1.0481
K, $m_{t\bar{t}} \in [345, 400]$	1.0000	1.0047	1.0148	1.0299	1.0429
K, $H_T \in [0, 90]$ GeV	1.0654	1.0656	1.0659	1.0661	1.0663
K, $p_T^t \in [0, 60]$ GeV	1.0403	1.0405	1.0408	1.0409	1.0411

($p^* < 100$, MC errors ≤ 0.0001.) In the first $m_{t\bar{t}}$ bin $K - 1$ grows by 0.2–0.3 percentage points per 1.25 GeV. The excess falls with m_t ($\Delta = 0.192 \to 0.164$ pb, -15%, from 170 to 175 GeV), but the LO rate in a bin with fixed edges falls faster as threshold moves towards its upper edge ($\approx 5.0 \to 3.4$ pb, -32%). A bin whose lower edge sits near $2m_t$ (345 GeV) is far more sensitive, since the toponium peak crosses the edge: K goes from 1 to 1.043. H_T, p_T^t and the rapidities barely move. Per-mass CSVs in the analysis binnings, with a README: `results/6_tfactors/toponium_kfactors_mtscan_13TeV/` (and `.zip`); all bins with an effect against m_t: `run7_mtscan_summary.txt`. → `6_tfactors.py mtscan`

7.11 Earlier full-sample runs

Before the window design, the T-factors were computed from pairs of independent full samples (10×10^6 events each, CT25NNLO, $t\bar{t} \to be^+\nu\,\bar{b}\mu^-\bar{\nu}$; window $W \leq 350$, $p^* \leq 50$): run 1, pp with the gg singlet modified only; run 2, gg colour singlet only; run 3, run 1 with the octet (G_8: coulomb_as25). Run 4 is run 3 with all e/μ combinations.

	run 1: pp, singlet	run 2: gg singlet	run 3: pp, $1 + 8$	run 4: pp, $1 + 8$, $\ell\ell$
$\sigma_{\mathrm{without}}$ [pb]	5.1229	0.9903	5.1229	20.1335
total $T - 1$	+0.92%	+4.68%	+0.79%	+0.77%
$m_{t\bar{t}} \in [250, 400]$ GeV	1.0457	1.209	1.0387	1.0375
$H_T \in [0, 90]$ GeV	1.0702	1.369	1.0601	1.0582

Run 3 against run 1 measures the octet. It changes the cross section by -0.0069 ± 0.0008 pb, i.e. -15% of the singlet's toponium excess (0.0473 pb), and lowers T by at most 0.010 (lowest H_T bin). Run 2 isolates the channel in which toponium forms: the same excess on a five times smaller denominator. Per-bin values and plots: `results/6_tfactors/run1_*/` to `run4_*/`.

Caveats.

1. LO matrix elements. K is a ratio meant to multiply higher-order $t\bar{t}$ predictions, assuming higher-order corrections largely cancel between numerator and denominator.
2. The window is a hard cut ($W \leq 350$ GeV, $p^* < 80$ for the factors, $p^* < p_{\rm cut}$ for the selection): the factors are switched off, not faded to 1, outside it. The choice of $p_{\rm cut}$ (50 or 100 GeV) moves Δ by 7%.
3. The octet potential is a model (Sec. 7.4); its four versions move K by at most 10^{-3}.
4. The window design agrees with full phase space once the $q\bar{q}$ samples hold both beam orderings (Sec. 7.8); a residual far below threshold ($W < 338$ GeV) in the full samples does not enter K.
5. One PDF (CT25NNLO) and one energy (13 TeV). K is a ratio, so the PDF enters mainly through the gg fraction at threshold and through α_s in G.

8 Where I fill in or depart from JKT

1. α_s is specified at m_Z and $\Lambda_{\overline{\rm MS}}$ obtained by solving (14) there; JKT quote α_s directly.
2. Grid: JKT's subintervals split once more, at 4 and 60 GeV (Step 5).
3. $\mathcal{A}(p)$ on a graded auxiliary Gauss–Legendre rule (Step 4); JKT only say "numerical integration".
4. K between the nodes by cubic-spline interpolation (Step 7); JKT do not say how they evaluate off the grid.
5. In the K-factor runs (Sec. 7) only, $\alpha_s(Q)$ in (13) is CT25NNLO's (LHAPDF), not (14).
6. The colour octet (Sec. 7.4): (13) with $C_F \to -1/6$ [HSY], α_s frozen below Q_0; [JKT] have no octet.

Everything else (the potential, the cut and its normalisation, the subtraction, the linear system) is [JKT] as printed.

A The code, file by file

Everything lives flat in `toponium_study/`. `toponium.py` is the only library; `0_...6_*.py` are the seven scripts you run (`uv run python 0_solver_checks.py`, and so on), each writing `results/<stage>/summary.txt` and figures. Every class opens with an "EQUATION MAP" listing which paper equation each method implements.

`toponium.py`: six classes, top to bottom

`Constants`
: m_t, Γ_t, C_F, n_F, and G_0, [JKT] (6).

`JKTPotential`
: Sec. 3: `alpha_2loop`, `lambda_msbar_2loop` (14); `V_pert` (13); `V_ir`, `continuity_constant_C`, `V_jkt` (15); `V_cut` (19); `V_cut_position`, `energy_shift` (20). `JKTPotential.coulomb(alpha)` gives fixed-coupling Coulomb with the same interface, to test the solver; `JKTPotential.octet_frozen(f, Q0)` the colour-octet potential of Sec. 7.4.

`LSSolver`
: Sec. 4: `kernel` (18), `jkt_grid` (23), `script_A` (22), `solve_K` (21), `G` (16). Usage: `LSSolver(potential).G(E, p)`.

`CoulombExact`
: Sec. 5.1: the exact Coulomb G (`G`, fast; `G_adaptive`, slow reference).

`FuksTables`
Fuks's swData and `CALCGREEN`, used as-is (reference only; nothing of mine is in it).

`Events`
the MadGraph events, read from `results/4_fuks_figs2to5/_cache_*.npz` (`read_lhe` takes $\ell = e, \mu$ on either side), and the weight $|G/G_0|^2$ for G from any of the above.

The seven stage scripts

script	what it does
`0_solver_checks.py`	solver vs exact Coulomb, convergence, $q_{\rm cut}$ independence (Sec. 5); `kernel_subtraction.png`
`1_validate_jkt.py`	solver vs JKT's own Figs. 2–5; `jkt_figs.png`
`2_identify_swdata.py`	which potential is in Fuks's tables (Sec. 6)
`3_fuks_fig1.py`	Fuks's Fig. 1 rebuilt from swData, my G alongside
`4_fuks_figs2to5.py`	events re-weighted with the three sources of G. Reads `_cache_full.npz`, `_cache_singlet330.npz`: **the only copy of the events, never delete them**. Also `--sample fuksproc_ct25` (Fuks's exact process, CT25NNLO) and `--octet` (their figures with the octet in the signal); the octet study with the Green's functions in MATRIX1 writes `fuks_octet_mg/`
`5_comparison_tables.py`	CSV tables for comparing with an independent solver (colour singlet): Λ, C, C_1, $\tilde{V}_0$ per setting; $\alpha_s(Q^2)$ and each piece of the potential vs Q^2; G on (E, p) grids
`6_tfactors.py`	toponium K-factors at 13 TeV (Sec. 7): LO MadGraph with the singlet and octet matrix elements multiplied by $\|G_1/G_0\|^2$, $\|G_8/G_0\|^2$ in MATRIX1. Here (13) uses CT25NNLO's own $\alpha_s(Q)$ (from LHAPDF) instead of (14), the same α_s MadGraph uses. Final run: `run6_final_kfactors_13TeV` (`setup`/`smoke`/`run`, then `kfactors`, `export` for the plain K-factor files, `arbiter` for Sec. 7.8; tables from `table run5`). Run 5, `run5_window_kfactors_13TeV`, is the student comparison (its $\mathrm{SM}_{\rm full}$ is run 4's). Earlier full-sample runs: `run1_pp_fullcolour_13TeV`, `run2_gg_singlet_13TeV`, `run3_pp_singlet_octet_13TeV`, `run4_pp_singlet_octet_ll_13TeV`. `octetcheck` (Sec. 7.7); the m_t scan `run7_mt*_13TeV` with `mtscan` (Sec. 7.10)

`reference/` holds the papers ([JKT], [Fuks], [HSY], and the CMS and ATLAS threshold measurements) and swData; `README.md` has the recipes to regenerate tables and samples; `pyproject.toml` and `uv.lock` define the Python 3.11 environment.